1. In your experience, how often do you find problems in an integration environment?
2. How often is your integration environment unstable?

In response to question 1, people rarely respond with *never* finding problems in the integration environment. However, most people respond by saying they *sometimes* find problems in the integration environment. In response to question 2, people rarely respond with *never* having stability problems in their integration environment. However, most people respond by *sometimes* having an unstable integration environment. The questions highlight both how important integration testing is and how inconsistent it is. Teams should also consider taking into account the time and cost of integration environments, as well as developer and tester time spent debugging.

12.1 When to use integration tests over contract tests

One of the most difficult parts of selling contract testing is persuading management and other people that contract tests add value beyond integration test coverage. We'll discuss identifying which integration tests to migrate in section 12.2. In many circumstances where legacy code exists, integration tests will already exist. In this scenario, there is an opportunity to consider which integration tests can be migrated to contract tests. We're not saying that all integration tests should be replaced by contract tests, and we'll look at refactoring legacy integration tests in section 12.3. There are other factors to consider when integrated environments are involved. For example, deployments are done via a build pipeline and infrastructure as code. Therefore, configuration is passed from build pipelines, and secrets are pulled from a secure vault. Then, these configuration values and secrets are passed to the infrastructure code. What could go wrong? It's easy for integration values to get missed in an environment or for variables to be referenced incorrectly. Integration tests catch these types of deployment problems, whereas contract tests only test at the boundary of the service or application, as shown in figure 12.2.